

Estimated Annual Agricultural Pesticide Use Grant County, Washington 2014-2018

24 July 2026

Summary

Grant County ranks **#1 of Washington’s 39 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of WA’s five highest pesticide-use counties. Grant County sits in the heart of the Columbia Basin Project, where large-scale center-pivot irrigation supports potatoes, field corn, alfalfa hay, and wheat.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Grant County, Washington** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Grant County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	3206654	3087616	2405936
2	DICHLOROPROPENE	OC	1988759	2023434	1821609
3	PETROLEUM OIL	OTH	972486	1264186	1158355
4	METAM POTASSIUM	DTC	832061	526477	509673
5	CALCIUM POLYSULFIDE	INO	468221	247948	206978
6	CHLOROPICRIN	OC	435327	178421	118628
7	SULFUR	INO	430645	341136	311769
8	BACILLUS AMYLOLIQUEFACIEN	MIC	260288	193315	127325
9	SULFURIC ACID	INO	205211	432092	434792
10	GLYPHOSATE	OTH	114274	127631	118724
11	COPPER	INO	84522	58631	45066
12	EPTC	TC	64061	67255	68604
13	PARAQUAT	DIP	62079	35931	30505
14	MANCOZEB	DTC	55544	73115	83654
15	CHLOROTHALONIL	OTH	37098	37086	35717
16	PENDIMETHALIN	OTH	29644	33682	30103
17	BACILLUS THURINGIENSIS	MIC	28849	13337	10464
18	COPPER SULFATE	INO	28068	19284	16186
19	GLUFOSINATE	OTH	27963	11682	9015
20	METOLACHLOR & METOLACHLOR-S	OTH	26662	23678	21858
21	METOLACHLOR-S	OTH	26662	23678	21641
22	OXAMYL	CB	24124	9596	16137
23	2,4-D	CPX	23316	23488	28182
24	DIMETHENAMID & DIMETHENAMID-P	OTH	21790	14911	11896
25	DIMETHENAMID-P	OTH	21790	14911	11896
26	METRIBUZIN	TRZ	21080	22717	20702
27	BURKHOLDERIA SPP	MIC	20505	15220	11486
28	COPPER HYDROXIDE	INO	18127	16699	14363
29	BENTAZONE	OTH	17478	19168	19780
30	THIOPHANATE-METHYL	OTH	17397	23459	27077
31	CHLORPYRIFOS	OP	14401	22141	20084
32	POTASSIUM BICARBONATE	INO	13442	12938	9178
33	ATRAZINE	TRZ	11894	13103	11902
34	DIMETHOATE	OP	11696	10038	9973
35	LINURON	OTH	10057	5471	3704
36	DIQUAT	DIP	9512	6609	6342
37	COPPER OXYCHLORIDE	INO	8875	5596	3878
38	MALEIC HYDRAZIDE	OTH	7871	8692	8751
39	CARBARYL	CB	7410	6536	12266
40	PENTHIOPYRAD	OTH	7220	4623	3612
41	BROMOXYNIL	OTH	6610	7711	10769
42	FLUAZINAM	OTH	6580	5141	5107
43	MCPA	CPX	6276	7207	8085
44	PHOSPHORIC ACID	INO	6154	5220	12188
45	DIURON	OTH	5983	4333	4961
46	METHOMYL	CB	5742	10659	8080

Table 2: Top 80 EPest-high Estimates, Grant County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	MEFENOXAM	OTH	5432	4523	3924
48	BOSCALID	OTH	5410	8525	8730
49	AUREOBASIDIUM PULLULANS	MIC	5353	3827	2320
50	ETHALFLURALIN	OTH	5320	8291	10146
51	KAOLIN CLAY	INO	5034	166764	118244
52	MCPB	CPX	5022	2596	1566
53	PROPICONAZOLE	OTH	5016	5554	4086
54	NORFLURAZON	OTH	4953	2762	3189
55	FLUTOLANIL	OTH	4260	1933	2435
56	SIMAZINE	TRZ	4201	1536	1593
57	SPIROMESIFEN	OTH	4002	3700	3433
58	TRIFLUMIZOLE	OTH	3952	4713	4360
59	HEXAZINONE	TRZ	3678	4554	4288
60	PYRACLOSTROBIN	OTH	3577	5017	4690
61	CLETHODIM	OTH	3369	2159	2070
62	SPIROTETRAMAT	OTH	3310	3061	2999
63	BUPROFEZIN	OTH	3127	2143	1840
64	PROPARGITE	OTH	3087	3010	4156
65	PROHEXADIONE	OTH	3078	1438	1801
66	CYMOXANIL	OTH	2849	2697	1921
67	FLUMIOXAZIN	OTH	2739	2313	1438
68	CYDIA POMONELLA	MIC	2565	870	527
69	ETHEPHON	OTH	2430	1509	3189
70	BIFENTHRIN	PYR	2201	3097	2242
71	BACILLUS SUBTILIS	MIC	2198	2111	1714
72	CHLORANTRANILIPROLE	OTH	2191	2143	2216
73	2,4-DB	CPX	2188	1712	1352
74	REYNOUTRIA SACHALINENSIS	MIC	2124	1708	1226
75	ACETAMIPRID	OTH	2116	1299	1100
76	PYMETROZINE	OTH	2089	1247	1299
77	IMIDACLOPRID	OTH	2054	2291	3252
78	AZOXYSTROBIN	OTH	2048	3726	3018
79	FLUOPYRAM	OTH	2039	2126	1656
80	FLUXAPYROXAD	OTH	2015	1711	1268

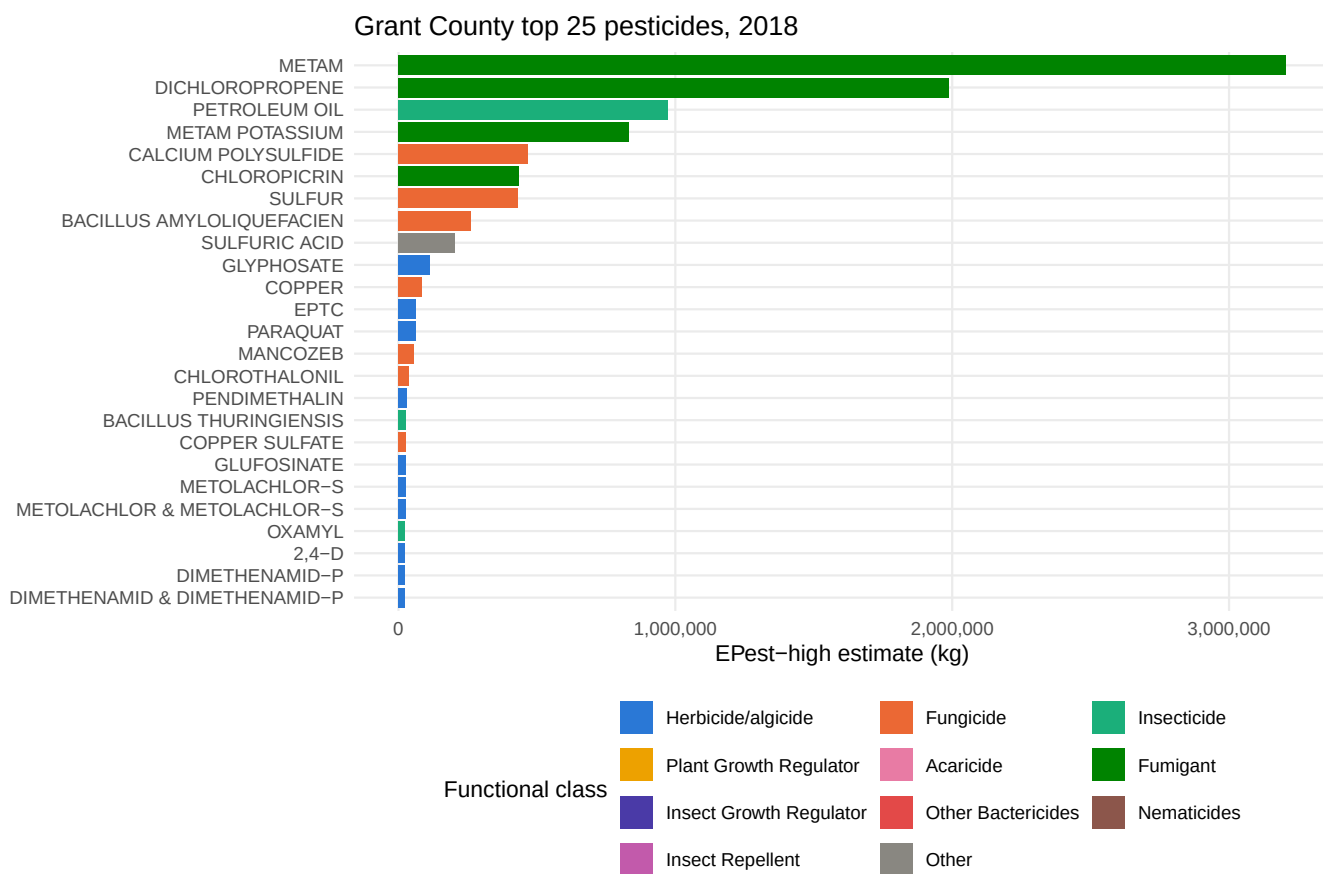


Figure 1: Top 25 compounds in Grant County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Grant County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	3206654	3206654
2	DICHLOROPROPENE	1988759	1988759
3	PETROLEUM OIL	969921	972486
4	METAM POTASSIUM	832061	832061
5	CALCIUM POLYSULFIDE	468210	468221
6	CHLOROPICRIN	0	435327
7	SULFUR	430410	430645
8	BACILLUS AMYLOLIQUEFACIEN	260288	260288
9	SULFURIC ACID	0	205211
10	GLYPHOSATE	110403	114274
11	COPPER	84361	84522
12	EPTC	64061	64061
13	PARAQUAT	61738	62079
14	MANCOZEB	53960	55544
15	CHLOROTHALONIL	37098	37098
16	PENDIMETHALIN	26652	29644
17	BACILLUS THURINGIENSIS	28846	28849
18	COPPER SULFATE	27639	28068
19	GLUFOSINATE	27963	27963
20	METOLACHLOR & METOLACHLOR-S	26662	26662
21	METOLACHLOR-S	26662	26662
22	OXAMYL	24124	24124
23	2,4-D	21926	23316
24	DIMETHENAMID & DIMETHENAMID-P	16690	21790
25	DIMETHENAMID-P	16690	21790
26	METRIBUZIN	18287	21080
27	BURKHOLDERIA SPP	16641	20505
28	COPPER HYDROXIDE	17865	18127
29	BENTAZONE	17478	17478
30	THIOPHANATE-METHYL	15482	17397
31	CHLORPYRIFOS	13598	14401
32	POTASSIUM BICARBONATE	198	13442
33	ATRAZINE	11894	11894
34	DIMETHOATE	11696	11696
35	LINURON	9868	10057
36	DIQUAT	9512	9512
37	COPPER OXYCHLORIDE	8875	8875
38	MALEIC HYDRAZIDE	7871	7871
39	CARBARYL	6935	7410
40	PENTHIOPYRAD	6249	7220
41	BROMOXYNIL	6610	6610
42	FLUAZINAM	6580	6580
43	MCPA	6276	6276
44	PHOSPHORIC ACID	6154	6154
45	DIURON	5882	5983
46	METHOMYL	5742	5742
47	MEFENOXAM	5062	5432
48	BOSCALID	5406	5410
49	AUREOBASIDIUM PULLULANS	5351	5353
50	ETHALFLURALIN	5320	5320

Table 4: Top 50 Grant County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	3087616	3087616
2	DICHLOROPROPENE	2023037	2023434
3	PETROLEUM OIL	1220856	1264186
4	METAM POTASSIUM	526477	526477
5	SULFURIC ACID	0	432092
6	SULFUR	337111	341136
7	CALCIUM POLYSULFIDE	247944	247948
8	BACILLUS AMYLOLIQUEFACIEN	193315	193315
9	CHLOROPICRIN	30791	178421
10	KAOLIN CLAY	11031	166764
11	GLYPHOSATE	118972	127631
12	MANCOZEB	72587	73115
13	EPTC	67255	67255
14	COPPER	58577	58631
15	CHLOROTHALONIL	37065	37086
16	PARAQUAT	34032	35931
17	PENDIMETHALIN	29941	33682
18	ZIRAM	0	24964
19	METOLACHLOR & METOLACHLOR-S	15046	23678
20	METOLACHLOR-S	15046	23678
21	2,4-D	21482	23488
22	THIOPHANATE-METHYL	21596	23459
23	METRIBUZIN	20977	22717
24	CHLORPYRIFOS	18859	22141
25	COPPER SULFATE	17488	19284
26	BENTAZONE	19168	19168
27	COPPER HYDROXIDE	15747	16699
28	BURKHOLDERIA SPP	13333	15220
29	DIMETHENAMID & DIMETHENAMID-P	12972	14911
30	DIMETHENAMID-P	12972	14911
31	BACILLUS THURINGIENSIS	13336	13337
32	ATRAZINE	13103	13103
33	POTASSIUM BICARBONATE	2787	12938
34	GLUFOSINATE	11174	11682
35	ETHOPROPHOS	11598	11598
36	METHOMYL	10659	10659
37	DIMETHOATE	9891	10038
38	OXAMYL	9197	9596
39	MALEIC HYDRAZIDE	8692	8692
40	BOSCALID	8524	8525
41	CUPROUS OXIDE	7984	8304
42	ETHALFLURALIN	8291	8291
43	DCPA	7987	7987
44	BROMOXYNIL	6449	7711
45	MCPA	5890	7207
46	DIQUAT	6609	6609
47	CARBARYL	6378	6536
48	COPPER OXYCHLORIDE	5595	5596
49	PROPICONAZOLE	4336	5554
50	LINURON	4690	5471

Table 5: Top 50 Grant County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	2405936	2405936
2	DICHLOROPROPENE	1819901	1821609
3	PETROLEUM OIL	1132357	1158355
4	METAM POTASSIUM	502646	509673
5	SULFURIC ACID	0	434792
6	SULFUR	308546	311769
7	CALCIUM POLYSULFIDE	206586	206978
8	BACILLUS AMYLOLIQUEFACIEN	127325	127325
9	GLYPHOSATE	112459	118724
10	CHLOROPICRIN	24918	118628
11	KAOLIN CLAY	24804	118244
12	MANCOZEB	82993	83654
13	EPTC	67751	68604
14	COPPER	44829	45066
15	CHLOROTHALONIL	35696	35717
16	PARAQUAT	28764	30505
17	PENDIMETHALIN	27212	30103
18	2,4-D	20496	28182
19	THIOPHANATE-METHYL	25392	27077
20	METOLACHLOR & METOLACHLOR-S	16456	21858
21	METOLACHLOR-S	16288	21641
22	METRIBUZIN	19036	20702
23	CHLORPYRIFOS	17652	20084
24	BENTAZONE	19780	19780
25	PINOLENE	17091	17091
26	COPPER SULFATE	12029	16186
27	OXAMYL	15897	16137
28	ZIRAM	2549	15031
29	COPPER HYDROXIDE	13549	14363
30	CARBARYL	12168	12266
31	PHOSPHORIC ACID	2157	12188
32	ATRAZINE	11902	11902
33	DIMETHENAMID & DIMETHENAMID-P	10350	11896
34	DIMETHENAMID-P	10350	11896
35	BURKHOLDERIA SPP	10000	11486
36	ENDOSULFAN	0	10829
37	BROMOXYNIL	6670	10769
38	BACILLUS THURINGIENSIS	10458	10464
39	ETHALFLURALIN	10146	10146
40	DIMETHOATE	9609	9973
41	POTASSIUM BICARBONATE	3087	9178
42	GLUFOSINATE	7434	9015
43	MALEIC HYDRAZIDE	8751	8751
44	BOSCALID	8712	8730
45	CUPROUS OXIDE	6068	8683
46	DCPA	5522	8105
47	MCPA	7292	8085
48	METHOMYL	7758	8080
49	ETHOPROPHOS	7661	7661
50	DIQUAT	6342	6342